

ORIGINAL INVESTIGATION



Bout duration and number of players of soccer small-sided games affect perceived enjoyment, physiological responses, and technical-tactical performance

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ABSTRACT

Purpose: To compare perceived enjoyment (PE), % of heart rate peak (%HRpeak), blood lactate (La), rating of perceived exertion (RPE), and technical-tactical performance of soccer players after four-a-side (4vs4) and three-a-side (3vs3) small-sided games (SSGs) with goalkeepers.

Methods: Sixteen males (20.7 ± 0.7 years; experience: 6.9 ± 1.2 years) completed and were measured after six sessions of SSGs (one twelve-minute bout (1x12), two six-minute bouts (2x6), and three four-minute bouts (3x4), both 3vs3 and 4vs4). Repeated measurements ANOVA evaluated significant differences ($p \leq 0.05$).

Results: There was higher PE for 1 × 12 compared to 2 × 6 (effect sizes [ES] = 1.53 [3vs3 and 4vs4]) and 3 × 4 (ES = 3.13 [4vs4] and 4.16 [3vs3]). Continuous bout duration (1x12) was characterized by a significantly greater percentage of successful passes compared to 2 × 6 (ES = 2.57 [4vs4] and 1.79 [3vs3]) and 3 × 4 (ES = 2.14 [4vs4] and 1.73 [3vs3]), and minor ball loss rate (2x6: ES = 2.14 [4vs4] and 1.11 [3vs3]; 3x4: ES = 2.11 [4vs4] and 1.57 [3vs3]). The percentage of successful tackles was only significantly greater for 1 × 12 in 4vs4 (ES = 0.83 [2x6] and 0.86 [3x4]) and successful duels in 3vs3 (ES = 1.41 [2x6] and 1.43 [3x4]). Significantly greater %HRpeak for 1 × 12 compared to 2 × 6 (ES = 1.02 [4vs4] and 0.81 [3vs3]) and 3 × 4 (ES = 1.04 [4vs4] and 0.99 [3vs3]) was observed, La differences were only significant in 3vs3 (ES = 1.60 [2x6] and 1.30 [3x4]). Greatest RPE were encountered in 3 × 4 (4vs4) and 1 × 12 and 3 × 4 (3vs3). Significant differences existed between the 3vs3 and 4vs4 designs with a non-clear pattern of better results for neither of both.

Conclusion: Therefore, coaches should consider longer continuous bouts when programming SSGs-based training to significantly increase PE, technical-tactical performance, and training load.

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KEYWORDS

Youth athletes; Physical Activity Enjoyment Scale (PACES); lactate; Rate of Perceived Exertion (RPE); heart rate percentage; YoYo test

Introduction

Small-sided games (SSGs) are multifunctional training exercises that allow team sport coaches to simultaneously enhance players' physical, technical, tactical, and psychological skills while being exposed to competitive match play scenarios (Dellal et al. 2012; Aguiar et al. 2015; Gonçalves et al. 2016; Moran et al. 2019). In the case of soccer, SSGs consist of playing soccer games (rules can be modified) in reduced spaces with less than 11 players, e.g., 3vs3 or 4vs4 (Hill-Haas et al. 2011; Köklü et al. 2017; Selmi et al. 2020). SSGs are an efficient way of training because they enable a coach to develop the skills and abilities of soccer players in simulated match environments (Hill-Haas et al. 2011), represent a time-efficient mode of periodizing the training in time-constrained environments, and are equally effective at enhancing some aspects of the performance to other methods (Moran et al. 2019). For instance, SSGs have been shown to induce similar physiological

responses compared to high-intensity interval training in professional soccer players (Selmi et al. 2018). Also, SSGs seem to be an effective way of maintaining player engagement through enhanced perceived enjoyment (Los Arcos et al. 2015; Selmi et al. 2018, 2020) and physical engagement (Hill-Haas et al. 2011). SSGs may develop individual and group emotional and social mental skills, leading to higher motivation, a more positive state of mind, greater physical enjoyment, and maintenance of players' training load (Los Arcos et al. 2015; Köklü et al. 2017; Selmi et al. 2018).

Manipulations of the conditions of SSGs (number of players, duration of the bout) are required to avoid training monotony and psychological tiredness, which may affect players' concentration and enjoyment levels during training (Monkam et al. 2008; Freitas et al. 2014; Clemente et al. 2016). Also, coach encouragement has been demonstrated to increase SSGs intensity (Rampinini et al. 2007). As a result, coaches need to

know how and when to appropriately structure SSGs training, in line with the stated goal of practice (Mallo and Navarro 2008). Previous studies have compared various programming variables relating to SSGs (Aguiar et al. 2015; Dellal et al. 2015). For instance, including SSGs at the beginning of the session may elicit longer distances run at different speed ranges (Sanchez-Sanchez et al. 2018). However, the choice of including SSGs at the beginning or end of the session, or one point or other of the weekly cycle, is influenced by coaches' preferences in terms of load regulation and/or technical-tactical learning (Rabbani et al. 2019). Due to all the aforementioned factors and given the particular importance of SSGs in soccer training, a deeper understanding of the physical, physiological, and psychological effects is required to properly manage and calibrate the training stimulus to drive optimal adaptation and engagement. To the best of our knowledge, no previous study has attempted to compare the effect of duration bout on the perceived enjoyment, training load, and technical-tactical responses to SSGs in soccer players.

To address this gap in the literature, this study was aimed at comparing the effects of different bout duration (1x12min, 2x6min, and 3x4min) and participant number (4vs4 and 3vs3) during SSGs on levels of perceived enjoyment, technical-tactical performance (% of successful passes, tackles, duels, and ball loss), and physiological responses (% of peak heart rate, rate of perceived exertion, and lactate) of soccer players. Our initial hypothesis was that the study dependent variables will be influenced by the SSGs bout duration and participant number, with continuous bouts offering higher physiological demands and more enjoyment, and a greater number of participants resulting in more conservative physiological responses with more technical-tactical actions.

Methods

Participants

Sixteen semi-professional male soccer players (age: 20.7 ± 0.7 years, height: 179.5 ± 6.1 cm, body mass: 67.2 ± 4.9 kg, body fat: $10.7 \pm 0.7\%$, soccer-playing experience: 6.9 ± 1.2 years; peak heart rate in the YoYo test: 187.1 ± 3.7 bpm) voluntarily participated in this study. All players competed for the same team at a national league in Tunisia (high-level). On average, the players trained 5 times per week (7.5 ± 1.54 hours of training/week), with one official competition. Goalkeepers were excluded from the analysis. Participants' level was characterized to avoid skill and fitness mismatches and a subsequent imbalance in opposing SSGs teams. Further details of this technique for group allocation named minimisation can be found in previous expert literature (Altman and Bland 2005). For such purpose, and bearing in mind that the talent selection should be done combining objectively measured data and coaches' assessments (Sieghartsleitner et al. 2019), the Yo-yo intermittent recovery test was carried out to obtain the participants' maximal heart rate. Also, the distance covered in this test was used as a classification criterion. The ranking system was as follows: players who covered the least distance were given a score of '1' and those who covered the greatest distance were given a

score of '5'. In addition, the coach of the team (being the coach of the team for 3 consecutive years), after brief instructions, provided a subjective rating of the overall technical-tactical skill level of each player using a 5-point scale (from 1 = poor to 5 = excellent). The total score for each player was then the sum of their technical-tactical skill and YYIRT scores. To avoid skill and fitness mismatches and subsequent imbalances in opposing SSGs teams, each SSGs team was then balanced using the total scores of the players involved.

The study was approved by the Ethics Committee of the University of Jendouba and was consistent with the institutional ethical requirements for human experimentation following the Declaration of Helsinki as it was revised in 2013. All the subjects or legal guardians (where applicable) were notified of the research procedures, requirements, benefits, and risks before giving written informed consent.

Procedures

This was a quasi-experimental study with a repeated measures design. Hence, the same players randomly performed the different SSGs conditions in separated days throughout three weeks mid-season, at the same time (between 2:00 and 3:00 p.m.), and on the same synthetic grass pitch. On Sundays, the players participated in a match, and Mondays were dedicated to recovery activities. A maximum of two SSGs sessions per week was undertaken. Measurements were taken during and after each SSGs to obtain data for the dependent variables (perceived enjoyment, technical-tactical performance, and psychophysiological responses).

During the two weeks before the intervention (a total of five weeks for the full study), all participants were familiarized with the testing procedures and SSGs formats. Also, anthropometric measures and the Yo-Yo test were performed in this period. Following the familiarization, participants performed the SSGs preceded by a 20-min specific warm-up, which consisted of low-intensity running, striding, and stretching. Players were allowed to consume water ad-libitum during recovery. Supervision and verbal encouragement were continuously provided by the coaches to encourage maximum performance (Rampinini et al. 2007) and spare balls were available to each team whenever the ball was out of bounds or when a goal was scored. SSGs were performed with goalkeepers, although they were not included in the study.

Small-Sided Games (SSGs)

The SSGs involved games with teams of 4vs4 and 3vs3 on two pitches sized 25×32 and 20×30 m respectively. The goals dimensions were 3 meters in length and 2 meters in height. The relative pitch size ($\text{m}^2/\text{number of players}$) for both designs (4v4 and 3vs3) was 100 m^2 as similarly performed by previous expert literature and bearing in mind that changes in the pitch size would entail changes in the training load and technical-tactical performance (Olthof et al. 2018; Clemente et al. 2021). The three different SSGs net durations were a) one bout of twelve minutes (1x12min), b) two bouts of six minutes (2x6min), and c) three bouts of four minutes (3x4min). HR was continuously monitored in five-second intervals throughout the SSGs. Perceived enjoyment and rate of perceived exertion

(RPE) were recorded at the end of each game. Passive recovery of two minutes was allowed between the intermittent SSGs bouts.

Anthropometric measurements

Measurements of body mass and height were performed as in previous research with young subjects (Hammami et al. 2021), using a height rod (IP0955; Invicta Plastics Limited, Leicester, England) and electronic scale (9179 SV3R; Salter, United Kingdom), respectively. Biceps, triceps, subscapular, and suprailiac skinfolds were measured in mm using a standard calliper (Baty International, West Sussex, England). Body density and percentage of body fat were calculated using the following formulas (Shephard et al. 1988).

Body density = $1.1765 - 0.0744 * (\log_{10} \Sigma S)$, where ΣS is the sum of the skinfolds.

Body fat (%) = $(4.95/D - 4.50) * 100$, where D is body density.

Yo-yo intermittent recovery test

The Yo-Yo Intermittent Recovery test (level one) was used to measure aerobic performance (Bangsbo et al. 2008) and characterize the sample in this regard. This protocol consists of repeated 20-m shuttle runs between the starting, turning, and finishing lines. The test continues at a progressively increasing speed which is controlled by audio bleeps from a recording. The speed level begins at 10 km/h. The test was stopped when a participant could no longer maintain the required running speed dictated by the bleep for two consecutive occasions, or if they felt that they could not complete the stage. The test was performed on a synthetic grass field in groups comprised of six players. The highest heart rate measurement (HRpeak) was recorded (see 'Participants' section). This test has proven useful to elicit HRpeak values very similar to the actual maximum heart rate (Coutts et al. 2009). Considering that the Yo-Yo test is commonly part of fitness assessment programs in soccer, all the participants were familiar with the testing procedures.

Physical Activity Enjoyment Scale (PACES)

Enjoyment of physical activity was assessed by the 18-item Physical Activity Enjoyment Scale (PACES) (Kendzierski & DeCarlo, 1991). All participants were familiarized with the use of the scale before rating how they 'feel at the moment about the physical activity you have just been doing', using a 7-point scale (e.g., 1 = 'It's very pleasant'; 7 = 'It's not fun at all'). Overall enjoyment of physical activity mean score was generated by summing the individual item means scores, yielding a potential score range from 18 to 126.

Technical-tactical analysis

Technical-tactical analysis data collection was performed using video cameras (Canon HF G40, Canon Inc, Tokyo, Japan). This camera allows unlimited access to the whole field and the observer may repeat or stop filming and allows him to count all technical-tactical parameters under investigation. The percentage of successful passes, duels, tackles, and ball loss were analysed from the total actions each player performed. These variables are included in previous research analysing the technical-tactical performance of soccer players (Přača et al. 2015; Yi et al. 2020). A successful pass was understood as a pass

reaching its initial aim and not being intercepted by a rival. Duels consisted of direct competitions between opposing players competing to win possession of the ball. Successful tackles were recognised as the act of a defender to meet an opponent who was in possession of the ball to legally take the ball away. And finally, the ball loss percentage was considered the number of ball possessions lost out of the total of balls touched. Goals were not considered as the goalkeepers were not classified by level. The same trained researcher (sports scientist specialized in soccer) analysed the videos twice (Sieghartsleitner et al. 2019).

Heart rate measurement

Heart rate was monitored using a Polar Team Sports System (Polar Electro Oy, Kempele, Finland). Stored data were transferred to a computer and filtered by Polar Precision Performance Software 4.0 (Polar Electro Oy, Kempele, Finland).

The percentage of the peak heart rate (%HRpeak) attained in each SSGs was used to assess cardiovascular acute adaptations to exercise. %HRpeak was calculated by dividing the peak heart rate of the SSGs (HRgame) by the peak heart rate (HRpeak; obtained in the Yo-Yo test) times 100; %HRpeak = $(HR_{\text{game}} / HR_{\text{peak}}) * 100$ (Dellal et al. 2008).

Blood lactate

Blood lactate (La) was measured during the 3–4 minutes after the end of the last bout of each SSGs as shown in the study of Taoutaou et al. (Taoutaou et al. 1996). The blood samples were collected at the fingertip of the index finger and were analysed by a portable lactate monitor (Lactate Pro, Arkray, Japan) which has been previously validated (Pyne et al. 2000).

Rate of Perceived Exertion (RPE)

After the last bout of each SSGs format, the perceived intensity of the training sessions for each player was evaluated using the Borg CR-10 Scale for perceived exertion as proposed by Foster et al. (Foster et al. 2001). Players were familiarized with the CR-10 scale. This scale has been validated as an indicator of training intensity in intermittent studies using SSGs in soccer (Coutts et al. 2009). The Yo-Yo test was also used to familiarize participants with what would be a 10 (maximum effort) in the perceived effort scale.

Statistical analysis

All statistical analyses were carried out using IBM SPSS Statistics for Windows, version 25.0 (IBM Corp, Armonk, NY). All data are expressed as mean $\pm$ standard deviation (SD). The cut-off criterion for statistical significance was uniformly set for the present study at $p \leq 0.05$.

The normality of the distribution for all variables was checked using the Kolmogorov-Smirnov test. All the dependent variables followed a normal Gaussian distribution. Therefore, an analysis of variance (ANOVA) with repeated measurements and two within-subject factors (bout duration and number of participants) was used to determine significance in the differences between SSGs on each dependent variable. Bonferroni correction was used for the comparisons. Effect size was evaluated with eta partial squared (ηp^2), where 0.01

$\eta^2 < 0.06$ constitutes a small effect, $0.06 \leq \eta^2 \leq 0.14$ constitutes a medium effect, and $\eta^2 > 0.14$ constitutes a large effect (Lakens 2013). Where significant differences were detected, post-hoc pairwise comparisons, also with Bonferroni corrections, identified where these differences occurred. The magnitude of the differences for the post-hoc comparisons was calculated as Cohen's d with Hedges' corrections (Lakens 2013). This value is reported as unbiased Cohen's d (d_{unb}) (Cumming 2014), with $d_{unb} < 0.50$, constituting a small effect, $0.50 \leq d_{unb} \leq 0.79$ moderate, and $d_{unb} \geq 0.80$ a large effect (Cohen 1988). Hereunder, d_{unb} will be named as d . 95% confidence intervals (95%CI) for the differences were also calculated (Batterham and Hopkins 2006; Hopkins 2007). Bearing in mind that a conservative correction (Bonferroni) for the statistical comparisons was used and the considerably large number of comparisons, special attention should be given to the effect sizes and confidence intervals when interpreting the differences presented to avoid dichotomy of the p -value and, therefore, type-I error (Rothman 2010; Greenland 2019).

Results

Table 1 presents the data on the perceived enjoyment, technical-tactical performance, and psychophysiological responses of each SSGs.

Perceived enjoyment

The ANOVA testing showed a significant interaction effect between both factors (bout duration and number of participants) on the perceived enjoyment, with $F(2,30) = 9.32$, $p = 0.001$, $\eta^2 = 0.38$. There was higher PE for 1×12 min compared to 2×6 min (4vs4: mean difference [m.d.] 8.19, 95%CI [6.19–10.19], $p < 0.001$, $d_{unb} = 1.53$; 3vs3: m.d. 9.44, 95%CI [4.23–14.65], $p = 0.001$, $d_{unb} = 1.53$) and 3×4 (m.d. 15.25, 95%CI [12.14–18.36], $p < 0.001$, $d_{unb} = 3.13$; 3vs3: m.d.

23.75, 95%CI [18.13–29.37], $p < 0.001$, $d_{unb} = 4.16$) in both 4vs4 and 3vs3. The full set of post-hoc pairwise comparisons can be found in Supplement 1.

Technical-tactical performance

The most notable findings were that a significant interaction between both factors was observed on the percentage of successful passes ($F(2, 30) = 3.83$, $p = 0.03$, $\eta^2 = 0.20$), tackles ($F(2, 30) = 5.35$, $p = 0.01$, $\eta^2 = 0.26$), and ball loss ($F(2,30) = 4.54$, $p = 0.02$, $\eta^2 = 0.23$). The interaction between the within-subject factors was not significant for the percentage of successful duels ($F(2, 30) = 0.20$, $p = 0.82$, $\eta^2 = 0.01$). Considering that both factors independently caused an effect on this variable (bout duration: $F(2, 30) = 12.28$, $p < 0.001$; $\eta^2 = 0.45$; number of participants: $F(1,15) = 1195.59$, $p < 0.001$, $\eta^2 = 0.99$) post-hoc analyses were also performed. Continuous bout duration (1×12) was characterized by a significantly greater percentage of successful passes compared to 2×6 (4vs4: m.d. 2.91, 95%CI [1.98–3.85], $p < 0.001$, $d_{unb} = 2.57$; 3vs3: m.d. 2.15, 95%CI [1.21–3.09], $p < 0.001$, $d_{unb} = 1.79$) and 3×4 (4vs4: m.d. 3.54, 95%CI [2.18–4.91], $p < 0.001$, $d_{unb} = 2.14$; 3vs3: m.d. 2.03, 95%CI [0.96–3.09], $p < 0.001$, $d_{unb} = 1.73$), and minor ball loss rate also compared to 2×6 min (4vs4: m.d. 2.44, 95%CI [1.61–3.26], $p < 0.001$, $d_{unb} = 2.14$; 3vs3: m.d. 1.19, 95%CI [0.51–1.87], $p = 0.006$, $d_{unb} = 1.11$) and 3×4 min (4vs4: m.d. 2.13, 95%CI [1.45–2.80], $p < 0.001$, $d_{unb} = 2.11$; 3vs3: m.d. 1.56, 95%CI [0.92–2.21], $p < 0.001$, $d_{unb} = 1.57$). The percentage of successful tackles was only significantly greater for 1×12 min in 4vs4 (comparison with 2×6 min: m.d. 8.80, 95%CI [1.76–15.85], $p = 0.05$, $d_{unb} = 0.83$; comparison with 3×4 min: m.d. 9.08, 95%CI [1.81–16.36], $p = 0.05$, $d_{unb} = 0.86$) and the percentage of successful duels in 3vs3 (comparison with 2×6 min: m.d. 5.92, 95%CI [3.16–8.68], $p = 0.001$, $d_{unb} = 1.41$; comparison with 3×4 min: m.d. 5.79, 95%CI [3.43–8.16], $p < 0.001$, $d_{unb} = 1.43$). The full set of post-hoc pairwise comparisons can be found in Supplement 2.

Table 1. Mean values $\pm$ SD of perceived enjoyment, technical-tactical performance, and psychophysiological responses during 4vs4 and 3vs3 soccer small-sided games (SSGs) at different bout duration (1×12 minutes, 2×6 minutes, and 3×4 minutes).

Dependent variables	Four-a side SSGs (4vs4)			Three-a side SSGs (3vs3)		
	Condition 1 (1×12 min)	Condition 2 (2×6 min)	Condition 3 (3×4 min)	Condition 4 (1×12 min)	Condition 5 (2×6 min)	Condition 6 (3×4 min)
PACES score	84.1 $\pm$ 5.2 ^{2,3,*}	75.9 $\pm$ 4.9 ^{3,*}	68.9 $\pm$ 4.0*	88.6 $\pm$ 5.1 ^{5,6}	79.1 $\pm$ 6.5 ⁶	64.8 $\pm$ 5.7
% of successful passes	60.1 $\pm$ 1.2 ^{2,3,*}	57.2 $\pm$ 0.9*	56.5 $\pm$ 1.9*	61.2 $\pm$ 1.2 ^{5,6}	59.1 $\pm$ 1.1	59.2 $\pm$ 1.0
% of successful tackles	78.5 $\pm$ 9.8 ^{2,3,*}	69.7 $\pm$ 10.4	69.4 $\pm$ 10.3	59.0 $\pm$ 13.3	63.6 $\pm$ 10.1	63.2 $\pm$ 12.8
% of successful duels	85.6 $\pm$ 5.6*	80.5 $\pm$ 4.8*	81.0 $\pm$ 5.3*	57.1 $\pm$ 4.6 ^{5,6}	51.2 $\pm$ 3.3	51.4 $\pm$ 2.9
% of ball loss	18.1 $\pm$ 0.9 ^{2,3,*}	20.5 $\pm$ 1.3*	20.2 $\pm$ 1.0*	16.9 $\pm$ 0.9 ^{5,6}	18.1 $\pm$ 0.9	18.5 $\pm$ 1.0
Peak HR of the game	166.7 $\pm$ 4.9	162.1 $\pm$ 2.0	161.6 $\pm$ 1.8	171.1 $\pm$ 4.7	167.0 $\pm$ 4.6	164.7 $\pm$ 5.0
% HRpeak	89.1 $\pm$ 2.7 ^{2,3,*}	86.7 $\pm$ 1.7*	86.4 $\pm$ 2.2*	91.5 $\pm$ 2.8 ^{2,3}	89.3 $\pm$ 2.3	88.1 $\pm$ 3.7
La (mmol/l)	7.0 $\pm$ 1.1*	6.4 $\pm$ 0.2 ^{3,*}	6.9 $\pm$ 0.3	8.2 $\pm$ 0.5 ^{5,6}	7.1 $\pm$ 0.7	7.3 $\pm$ 0.8
RPE	4.2 $\pm$ 1.2 ^{3,*}	3.6 $\pm$ 0.6 ^{3,*}	5.1 $\pm$ 0.9	5.9 $\pm$ 0.8 ⁵	4.6 $\pm$ 0.5 ⁶	5.4 $\pm$ 0.8

*: Statistically significant difference ($p < 0.05$) between 4vs4 and 3vs3; ^{1,2,3,4,5,6}: statistically significant difference with Condition 1,2,3,4,5, or 6, respectively. PACES: Physical Activity Enjoyment Scale; %HRpeak: percentage of the peak heart rate achieved; RPE: rate of perceived exertion measured with the Borg CR-10 scale; La: lactate.

Psychophysiological responses

A significant interaction existed between the bout duration and the number of participants for RPE ($F(2,30) = 7.27, p = 0.003, \eta^2 = 0.33$) and La responses ($F(2,30) = 3.23, p = 0.05, \eta^2 = 0.18$). The interaction between bout duration and number of participants resulted non-significant for the %HRpeak ($F(2,30) = 0.59, p = 0.56, \eta^2 = 0.04$). Considering the similar relative pitch size for different number of participants, this result may imply that the effects of bout duration on the %HRpeak did not depend on the number of participants. Bearing in mind that both factors showed a significant independent effect on this variable (bout duration: $F(2,30) = 20.53, p < 0.001, \eta^2 = 0.58$; number of participants: $F(1,15) = 18.52, p = 0.001, \eta^2 = 0.55$) post-hoc analyses were also performed for this dependent variable. Significantly greater %HRpeak for 1×12 min compared to 2×6 (4vs4: m.d. 2.47, 95%CI [1.23–3.71], $p = 0.001, d_{unb} = 1.02$; 3vs3: m.d. 2.22, 95%CI [0.78–3.66], $p = 0.015, d_{unb} = 0.81$) and 3×4 min (4vs4: m.d. 2.72, 95%CI [1.22–4.22], $p = 0.002, d_{unb} = 1.04$; 3vs3: m.d. 3.42, 95%CI [2.08–4.76], $p < 0.001, d_{unb} = 0.99$) was observed. La differences were only significant in 3vs3 (comparison with 2×6 min: m.d. 1.09 mmol/l, 95%CI [0.58–1.59], $p < 0.001, d_{unb} = 1.60$; comparison with 3×4 min: m.d. 0.91 mmol/l, 95%CI [0.50–1.31], $p < 0.001, d_{unb} = 1.30$). Greatest RPE were encountered in the 3×4 min bout for the 4vs4 design (comparison with 1×12 min: m.d. 0.88, 95%CI [0.03–1.72], $p = 0.04, d_{unb} = 0.81$; comparison with 2×6 min: m.d. 1.50, 95%CI [0.65–2.35], $p = 0.001, d_{unb} = 1.91$) and in both the 1×12 min and 3×4 min for the 3vs3 design (1×12 min compared to 2×6 min: m.d. 1.38, 95%CI [0.60–2.15], $p = 0.001, d_{unb} = 2.10$; 3×4 min compared to 2×6 min: m.d. 0.81, 95%CI [0.07–1.56], $p = 0.03, d_{unb} = 1.21$). The full set of post-hoc pairwise comparisons can be found in Supplement 3.

Discussion

The purpose of this study was to assess differences in levels of perceived enjoyment, technical-tactical performance, and physiological responses between three bout durations (1×12 min, 2×6 min, and 3×4 min) during 4vs4 and 3vs3 SSGs. The main finding of this investigation was that a longer bout duration (in this study one bout of 12 minutes) has a positive influence on perceived enjoyment and technical-tactical performance. Regarding the physiological responses to the 1×12 min bout, it is worth highlighting a significantly higher %HRpeak, non-significantly higher La concentrations (significant in the 3vs3 design), significantly lower RPE compared with the 3×4 min bout in the 4vs4 design (non-significant differences with the 2×6 min bout), and significantly higher RPE compared to 2×6 min bout in the 3vs3 design (non-significant compared to 3×4 min bout). While caution should be applied since we did not measure the adaptations in the long term, the results suggest that coaches should consider the duration of SSGs when planning SSGs training and that one longer bout should be chosen over many short bouts, to engage participants and obtain higher levels of physiological and technical-tactical performance in the short term. It is also worth highlighting that, participants reported enjoying more the 3vs3 games compared to 4vs4 in most of the time bouts.

To our knowledge, no previous study has investigated the relationship between perceived enjoyment and SSGs bout duration. Coaches should consider the importance of psychological factors in SSGs as it seems to affect the levels of intrinsic motivation of players (Selmi et al. 2018). In general, SSGs induce positive psychological reactions in players by motivating them to perform for long periods (Rowan et al. 2012; Carraro et al. 2014). The tendency of coaches to choose the longest bout duration in training seems to be linked to the phenomena of self-determination and intrinsic motivation (Chitwood et al. 1997). In our study, the longest bout duration provided the most interesting outcomes in terms of higher perceived enjoyment and technical-tactical performance. Concerning the physiological outcomes, the longest bout duration could be chosen when aiming to achieve higher load responses and intermittent bouts when lower intensity is the goal. This may be due to the fact that a long bout may resemble better the real activity of football than several short bouts, which may motivate players to strive and achieve a positive emotion of enjoyment (Los Arcos et al. 2015). Previous studies (Kimićek and Harris 1996; Alvarez et al. 2009) have shown that among the players who have the highest levels of satisfaction and motivation in SSGs, are those who enjoy sports most and report being more need intrinsically satisfaction. As such, SSGs training may provide a stimulus of potential psychological benefits and provide more enjoyment that can help coaches to better manage players during the season. Similarly, one study (Los Arcos et al. 2015) showed that during the last weeks of the season, SSGs training in addition to the training of the technical and tactical aspects maintains cardiorespiratory fitness and promotes a high level of enjoyment in youth elite soccer players. Although the present study showed greater enjoyment levels in the longest bout duration (see Table 1), other studies are underway to determine the optimal intensities of the SSGs in this aspect. Furthermore, other subjective experiences such as the rate of fatigue, feeling scale, and other questionnaires and/or semi-structured interviews should be included in future research.

The video analysis showed that bout duration during 4vs4 and 3vs3 SSGs influenced the technical-tactical performance (see Table 1). Specifically, our results revealed that players seemed to be less technically proficient during shorter interval bouts than continuous bouts. Indeed, all the technical-tactical variables were enhanced during the continuous bout compared to the interval format of SSGs (2×6 min and 3×4 min) being all of them of equal total duration. This may be due to increased perceived enjoyment, which may entail better technical-tactical performance. This rationale coincides with the findings of previous studies that measured enjoyment and mood responses to SSGs (Los Arcos et al. 2015; Selmi et al. 2018). On the specific influence of the number of participants on the technical-tactical performance, it is worth highlighting that slightly worse, although significant, percentages of successful passes and ball loss were obtained in 4vs4 compared to 3vs3, and better successful tackles and duels. These results may be understood through the higher number of rivals, which provide the game with greater uncertainty and therefore increases the chances of failure (in our study lower percentage of successful passes and a greater percentage of ball loss).

Concerning physiological responses, the present study indicates that SSGs bout duration influences %HRpeak, La, and RPE. Previous research (Aguilar et al. 2015; Köklü et al. 2017) suggested that the variation of bout duration modifies physiological responses and game behaviour in soccer players, which coincides with our results. Increasing the duration of SSGs bouts to 12 minutes resulted in an augmented cardiovascular effort, La, and RPE responses. Specifically, the 1x12min condition induced an intensity of ~90% of peak HR that is similar to the intensity achieved in a soccer match (Esposito et al. 2004). Such values are considered to be an adequate stimulus for the cardiovascular system (Hoff et al. 2002; Hulka et al. 2016; Köklü et al. 2017). Physiological responses during 1 × 12 minutes of SSGs in terms of La and RPE were relatively higher compared to the interval-type SSGs (2x6 and 3 × 4 min). These dissimilarities could be caused by reduced rest and differences in the number of technical-tactical actions such as the ball contacts. Our results are supported by previous works (Dellal et al. 2008), who studied different SSGs protocols (2vs2, 3vs3, 4vs4, 5vs5) and found a rise of 10.7% in the mean HR in elite soccer players. In the same context, one study (Köklü et al. 2017) stated that the use of SSGs of continuous bouts (e.g., 1x12min) increased HR, La, and RPE during 2vs2, 3vs3, and 4vs4 SSGs. Likewise, our results show that a longer bout duration causes a greater cardiovascular demand than a short bout duration in SSGs. Accordingly, continuous bout durations of 4vs4 and 3vs3 SSGs may be used to develop aerobic performance (Casamichana and Castellano 2010; Hulka et al. 2016; Köklü et al. 2017). Regarding the specific influence of the number of players on the physiological variables, it is worth highlighting that three-a-side SSGs induce higher cardiovascular and lactate requirements and provoke a higher rate of perceived exertion than four-a-side games.

Limitations and future directions

Although all the procedures have been carefully designed and supervised by trained researchers, several limitations should be listed.

First, it is worth highlighting that although previous expert literature highlights the validity of the coach in identifying talent (Sieghartsleitner et al. 2019), future studies with similar procedures as in the present study, should evaluate the validity of the coaches in organizing groups of the same level and identifying technical-tactical performance. While we controlled the participants' level in terms of YoYo test performance and technical-tactical assessment of the coach, other physiological and/or performance parameters could have been used. Also, players' fitness was not assessed post-training to account for fitness changes that could have occurred during the study. Therefore, future studies should check for long-term adaptations with SSGs to be able to prescribe the exact dose of SSGs for optimal adaptations. Finally, another aspect that could be addressed in future studies is the familiarisation procedures underwent by the participants in terms of the RPE (Borg and Love 2017), differentiating between the understanding of the scales by each player (Winter 2005; Macpherson et al. 2019).

Practical applications

This study showed that SSGs bout duration and the number of participants influenced perceived enjoyment, technical-tactical performance, and physiological responses of soccer players, favouring a 1 × 12 minute continuous format over 2x6min or 3x4min interval-type formats. From a practical point of view, coaches should consider longer continuous bouts when programming SSGs-based training to significantly increase perceived enjoyment, technical-tactical competence, and training load, due to provoking higher heart rate peak and lactate concentrations.

In regard to the number of participants, coaches should consider the results presented in this study. If a considerably high heart rate peak and lactate concentration (also a higher RPE) is needed due to the programming requirements, a 3vs3 design could be chosen over 4vs4, also bearing in mind that participants further enjoyed this SSGs design. Concerning technical-tactical actions both SSGs designs studied (3vs3 and 4vs4) could be considered and exchangeability applied to enhance the performance.

Summary

As far as we are aware, this is the first study to investigate the effects of different bout durations (12 minutes, 6 minutes, and 3 minutes) and formats (continuous and intervals) on the perceived enjoyment, technical-tactical performance, and physiological adaptations during small-sided games (SSGs). Also, three-a-side and four-a-side SSGs were compared. The results presented in this study highlight the need for controlling the bout duration due to presenting significant differences in players' engagement, technical-tactical performance, and physiological acute adaptations, which may be interrelated and enhance training load and competitive performance. In summary, better results were obtained for the longest continuous bout SSGs design (one bout of twelve minutes) and participants preferred the three-a-side design. The results presented in this study could be of great interest for coaches to be interpreted and applied depending on the requirements of each moment of the season.

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